Ignacio Vigil Cardenal

+34 673231760 | ivigilcardenal@gmail.com | linkedin.com/in/ignacio-vigil-cardenal

EDUCATION

University of Warwick **Warwick, UK**
BSc Mathematics and Statistics Hons (First Year Grade: ***89%*** */ Expected First Class 1:1)* *Sep 2024 - July 2027*

- Completed Modules: Linear Statistical Modelling (**R**), Algorithms, PDEs, Numerical Methods, Probability and Mathematical Statistics.
- Upcoming Modules (Year 3): Mathematical Finance, Time Series, Bayesian Forecasting, Stochastic Processes, Machine Learning.

University of Oxford (Lady Margaret Hall) **Online**
Course: Artificial Intelligence and Machine Learning: Theory and Practice ***(99,2%)*** *July 2025*

- Learned topics such as supervised learning (linear and logistic regression), neural networks and convolutional neural networks (CNNs).
- Engineered a multi-modal sentiment analysis model with **PyTorch** by fusing a CNN and an LSTM on a dataset of tweets.

International College Spain **Madrid, Spain**
International Baccalaureate (Bilingual Diploma): Achieved 42/45 *Sep 2022 - July 2024*

- Mathematics AA (7) (Best Grade in Cohort), Physics (7), Economics (7), Geography, Spanish, English.

PROFESSIONAL EXPERIENCE

KPMG **Madrid, Spain**
Quantitative Risk Management Intern *June 2026 – August 2026*

- Engineered Value at Risk (VaR) and Expected Shortfall models to quantify market risk using time series analysis and cointegration.
- Evaluated asset price stationarity and credit risk exposures using ADF tests to enhance systematic risk mitigation.

Undergraduate Research Support Scheme (URSS) by the University of Warwick, **Warwick, UK**
Researcher in the Department of Statistics *June 2025 –September 2025*
Project: Machine Learning for Brain Tumour Detection

- Engineered and evaluated machine learning and deep learning models to detect brain tumours from MRI scans using **PyTorch**.
- Implemented convolutional neural networks (CNNs) and Bayesian CNNs with PyTorch, achieving over 99% accuracy on the test set.

EXTRA-CURRICULAR ACTIVITIES

Warwick Business School Society **Warwick, UK**
Head of Quantitative Investment Group *Sep 2025 – Present*

- Designed a course on machine learning for quantitative investing, covering supervised learning methods, NLP and time series.
- Incorporated real-world financial data for portfolio optimisation and financial case studies to enhance members' practical skills.

IMC Prosperity 3-4 Challenge **UK**
Participant *March 2025 – April 2026*

- Ranked 3rd nationally, developing market-making and mean-reversion algorithms while executing basket trading and options pricing.

Warwick Mathematical Finance Magazine **Warwick, UK**
Contributor *Sep 2024 - Present*

- Applied Monte Carlo simulations to price European long call options with Python while explaining the
- Evaluated statistical arbitrage strategies, leveraging econometric analysis with systematic trading, yielding 15%+ annual returns.

SELECTED INDEPENDENT PROJECTS

Cross-Asset Trend Following

- Engineered a systematic macro trend-following strategy across equities, bonds, FX, and commodities using time series momentum signals, such as EWMA crossover, incorporating volatility-targeting position sizing and out-of-sample walk-forward validation.

Bayesian Factor Model for FX Carry

- Implemented a Bayesian linear factor model to decompose FX returns, utilising interest rate differential (carry) as the primary factor alongside momentum and value signals, evaluating posterior uncertainty to generate robust out-of-sample predictive distributions.

SKILLS, INTERESTS & CERTIFICATES

Technical skills: Python (Pandas, NumPy, PyTorch, Matplotlib, Scikit-Learn), Java, R, SQL, Excel and Microsoft Office
Languages: Spanish (Native), English (Fluent)
Certificates: Harvard's CS50 (C, SQL and Python), Kaggle Intro to Machine Learning, IBM Intro to Computer Vision and Image Processing
Interests: Basketball, Football, Gym, Padel, Running, Skiing, Scuba Diving, Music